Amartya Dutta

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Research Focus

My research focuses on drug repurposing using graph-based and large language model (LLM)–driven approaches through heterogeneous biomedical graphs and multimodal reasoning. I am also interested in advancing scene graph understanding using multimodal language models, particularly for open-world, zero-shot visual reasoning.

Education

Virginia Tech, Blacksburg, USA
PhD, Computer Science; GPA: 3.91/4 — May 2025 – Present
MS, Computer Science; GPA: 3.9/4 — Aug 2022 – Feb 2025
Thesis: Zero-Shot Scene Graph Relationship Prediction using VLMs

Indian Institute of Information Technology (IIIT) Guwahati, India
Bachelor of Technology, Computer Science and Engineering; GPA: 3.44/4 — Jul 2017 – May 2021

Work Experience

Graduate Research Assistant, COMPASS Centre — May 2025 – Present

- Researching host-centric, graph-based approaches for antiviral drug repurposing and developing deep learning models to uncover novel virus–drug relationships to accelerate therapeutic discovery.
- Contributed to a virology-focused scientific information extraction project by benchmarking a curated virus-mutation dataset against state-of-the-art LLM-based RAG frameworks to assess extraction performance on domain-specific biomedical literature.

Graduate Research Assistant, Virginia Tech — Aug 2022 – Dec 2024

- Developed a **novel zero-shot approach** for **Scene Graph Relationship Prediction** using **VLMs**, reframing Predicate Classification as an **MCQA task** and surpassing trained baselines by at least 7% for **balanced relationship prediction**. Designed an **open-ended relationship generation framework** to eliminate answer-choice biases and improve contextual understanding.
- Performed comparative analysis of **Weakly Supervised Semantic Segmentation** approaches, highlighting the superiority of saliency maps over CAMs and introducing stochastic aggregation to enhance saliency effectiveness.

Augmented Reality Developer Intern, Amply — Dec 2019 – Mar 2020

- Developed interactive Augmented Reality (AR) portals using AR-Core in Unity3D, enabling secure interactions with virtual objects to create immersive AR tours for client companies.

Virtual Reality Developer Intern, IIT Guwahati — May 2019 – Jul 2019

- Designed and developed an interactive Virtual Reality (VR) tour using Unity3D, focusing on smooth navigation within the virtual environment to enhance user experience (UX) in VR.

Publications

1. Blessy Antony, **Amartya Dutta**, Anuj Karpatne, T. Murali et al. “VILLA: Versatile Information retrieval from scientific Literature using large LAnguage models”. Under Review.
2. **Amartya Dutta**, M. Maruf, Ismini Lourentzou, Arka Daw, Anuj Karpatne et al. “Open World Scene Graph Generation using Vision Language Models”. CVPR 2025 Workshop, ICML 2025 Workshop. Paper
3. Abhilash Neog, Medha Sawhney, Kazi Sajeed Mehrab, **Amartya Dutta**, Anuj Karpatne et al. “Toward Scientific Foundation Models for Aquatic Ecosystems”. ICML 2025 Workshop
4. Medha Sawhney, Abhilash Neog, Mridul Khurana, **Amartya Dutta**, Arka Daw, Anuj Karpatne. “Physics-guided Diffusion Neural Operators for Solving Forward and Inverse PDEs”. CVPR 2025 Workshop Poster
5. Sepideh Fatemi, Abhilash Neog, Emma Marchisin, **Amartya Dutta**, Anuj Karpatne et al. “Scientific Equation Discovery using Modular Symbolic Regression via Vision-Language Guidance”. CVPR 2025 Workshop Poster
6. M. Maruf, Arka Daw, **Amartya Dutta**, Jie Bu, Anuj Karpatne. “Beyond Discriminative Regions: Saliency Maps as Alternatives to CAMs.” Paper
7. **Amartya Dutta**, Rajat Kanti Bhattacharjee, Ferdous Ahmed Barbhuiya. “Efficient Detection of Lesions During Endoscopy.” ICPR Workshop 2021. Paper
8. **Amartya Dutta**, Kamaljyoti Nath. “Learning via LSTM for Railway Bridge strains.” ICDSMLA 2020. Paper
9. **Amartya Dutta**, Ferdous Ahmed Barbhuiya. “Predicting Popularity of Images Over 30 Days.” Paper

Projects

- **SealQA:** Contributed towards the creation a dataset of complex questions that challenge state-of-the-art LLMs. This project evaluates how these models handle queries requiring up-to-date knowledge and complex reasoning by leveraging search engines for real-time information. Paper

- **Evaluating Model Reasoning and Hallucinations in Medical LLMs:** This project investigates factual error propagation in open-source medical LLMs (e.g., BioMistral, Asclepius) and documents their datasets for transparency. By highlighting performance variations, it aims to guide the development of safer, more reliable language models for healthcare. GitHub

- **Visualizing the Spotify Soundscape:** This project visualizes the Spotify Top 50 Tracks of 2023 through an interactive, HTML-based dashboard. Using D3.js and Plotly.js, it enables dynamic, data-driven exploration of each track's popularity and attributes. GitHub

- **Predicting Popularity of Flickr Images (ICIP 2021):** This project predicts how popular a Flickr image will be over 30 days, even before it's uploaded. By analyzing user and image social features alongside image visuals, it models engagement based on two factors: scale and shape. Using these factors, the method forecasts the daily engagement sequence. GitHub

Teaching Experience

- **CS 2064:** Intermediate Programming in Python – Graduate Teaching Assistant, Virginia Tech
- **CS 5644:** Machine Learning with Big Data – Graduate Teaching Assistant, Virginia Tech
- **CS 5805:** Machine Learning – Graduate Teaching Assistant, Virginia Tech

Academic Service & Awards

- **Reviewer:** KDD 2026: AI4Sciences Track
- **Reviewer:** International Journal of Computer Vision (IJCV) 2025, 2026
- **Reviewer:** CVPR 2025 Workshop, Computer Vision in the Wild (CVinW)
- **Reviewer:** AAAI-24-Imageomics-Workshop
- **Program Committee:** CVPR 2025 Workshop, CVinW
- **3rd Position:** IEEE ICIP Image Popularity Prediction Challenge (Oct 2020) View Results

Skills Summary

- **Languages:** Python, LaTeX, C++, Shell
- **Frameworks/Tools:** PyTorch, TensorFlow, Keras, Scikit, OpenCV, Numpy, Pandas, Git, Unity3D, vLLM, LangChain, FAISS, OpenAI API, Hugging Face Transformers & APIs